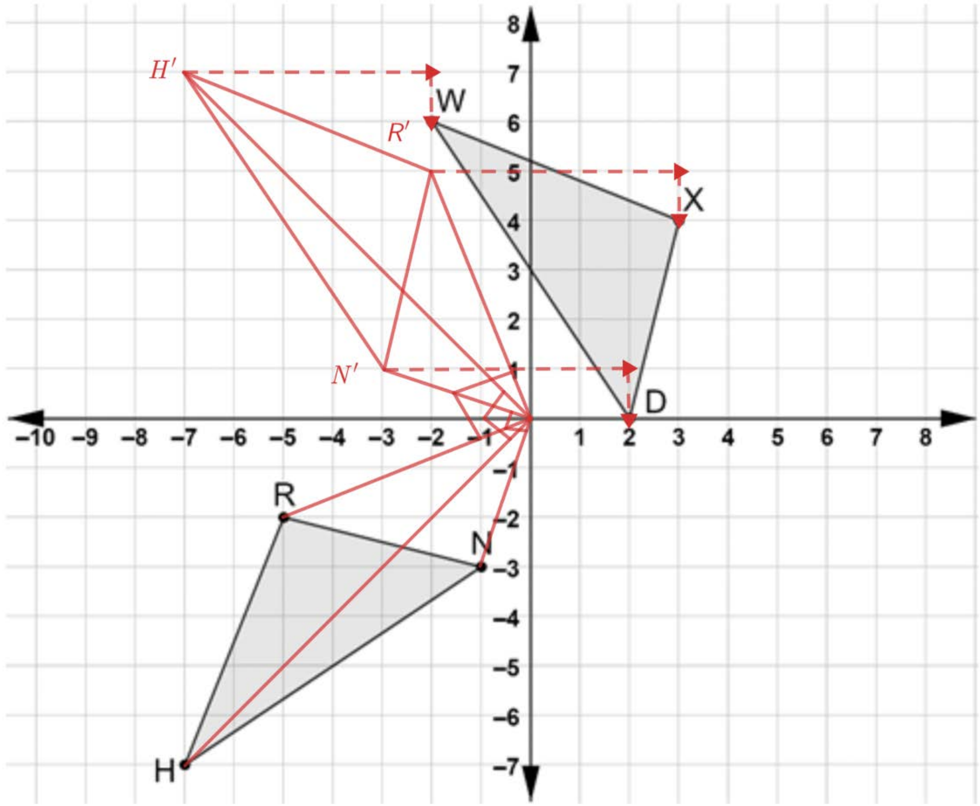
Geometry | Name___Answer Key_______
Unit 5 | Date____________________
Lesson H1 Homework | Period___________________

Triangles WXD and RNH are shown. Assume $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, and $\overline{DW} \cong \overline{NH}$.



1. Explain how to use geometric tools to verify that $\triangle HRN \cong \triangle WXD$.

   Use a ruler to measure all corresponding sides.

2. Write a sequence of transformations that will map $\triangle HRN$ onto $\triangle WXD$.

   Rotate $\triangle HRN$ $90°$ clockwise about the origin, then translate 5 units right and 1 unit down. (Answers may vary)

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'R'N'$ the result of the first transformation.

4. Map the vertices of $\triangle HRN$ to $\triangle H'R'N'$.

5. Map the vertices of $\triangle H'R'N'$ to $\triangle WXD$.

Answers may vary, but should align with #2.

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HRN$ to $\triangle H'R'N'$ and then from $\triangle H'R'N'$ to $\triangle$ _WXD_ shows that point H' coincides with point _W_____, point R' coincides with point ___X____, and point N' coincides with point ___D____.

7. Because the definition of ____congruence____ in terms of ____rigid motions____ preserves ________distance________, it can be stated that $\triangle HRN \cong \triangle$ _WXD_.

8. Since it was assumed that $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, and $\overline{DW} \cong \overline{NH}$ then _SSS____________ is sufficient to show triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{WX} \cong \overline{HR},\ \overline{RN} \cong \overline{XD},\ \angle RNH \cong \angle XDW$
- ☐ $\overline{RN} \cong \overline{XD},\ \overline{DW} \cong \overline{NH},\ \angle RNH \cong \angle XDW$
- ☐ $\overline{WX} \cong \overline{HR},\ \overline{DW} \cong \overline{NH},\ \angle RHN \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR},\ \overline{RN} \cong \overline{XD},\ \angle NHR \cong \angle XWD$
- ☐ $\overline{RN} \cong \overline{XD},\ \overline{DW} \cong \overline{NH},\ \angle NHR \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR},\ \overline{DW} \cong \overline{NH},\ \angle NHR \cong \angle XWD$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{WX} \cong \overline{HR},\ \angle RNH \cong \angle XDW,\ \angle RHN \cong \angle XWD$
- ☐ $\overline{DW} \cong \overline{NH},\ \angle RNH \cong \angle XDW,\ \angle NHR \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR},\ \angle RHN \cong \angle XWD,\ \angle NRH \cong \angle DXW$
- ☐ $\overline{RN} \cong \overline{XD},\ \angle RNH \cong \angle XDW,\ \angle NRH \cong \angle DXW$
- ☐ $\overline{DW} \cong \overline{NH},\ \angle NHR \cong \angle XWD,\ \angle NRH \cong \angle DXW$
- ☐ $\overline{RN} \cong \overline{XD},\ \angle NHR \cong \angle XWD,\ \angle NRH \cong \angle DXW$